



Artificial Intelligence

Department of Computer Science

University of Engineering and Technology, Lahore



Class Learning Outcomes:

Students shall be able to learn

- Control Structures
 - Sequential Structure
 - Decision/Selection Structure
 - if statement
 - if/else statement
 - if/elif/else statement
 - Repetition Structure
 - while structure
 - for structure

CLO2

Practice Programs

Write the following program in your jupyter notebook and state the output in the space provided.

Program to add two numbers

0.5

```
# Program to add 2 numbers
```

```
num1 = 5  
num2 = 10  
sum = num1 + num2
```

```
print("Number = " , num1)  
print("Number = " , num2)  
print("Sum = " , sum)
```

```
Number = 5  
Number = 10  
Sum = 15
```

```
# Program to add 2 numbers
```

```
num1 = 5  
num2 = 10  
sum = num1 + num2
```

```
print("Number = " + str(num1))  
print("Number = " + str(num2))  
print("Sum = " + str(sum))
```

Program to find velocity of a moving object

0.5

```
# Program to find velocity of a moving object
initialVelocity = float(input("Enter the initial velocity of the moving object "))

# acceleration in m per second square
acceleration = float(input("Enter the acceleration of the moving object "))

# time taken in second
time = float(input("Enter the time of the moving object "))

finalVelocity = initialVelocity + acceleration * time

print("Final velocity = ", finalVelocity)
```

Program to find Subject Status

1

```
# Program to update the subject status to student
Grade = int(input("Enter your marks: "))
Attendance = float(input("Enter your attendance percentage: "))

if Grade > 50 and Attendance > 75:
    print("Pass!")
elif Grade > 50 and Attendance >= 50:
    print("Pass!")
else:
    print ("Fail!")
```

What's the output with the following input?

Grade = 60 Attendance = 40	Grade = 70 Attendance = 75	Grade = 30 Attendance = 95

Program to find mess menu

1

```
# Program to find the mess menu
Day = str(input("Enter the day: "))

if Day == "Monday" or Day == "monday":
    print("Chicken Karahi")
elif Day == "Tuesday" or Day == "tuesday":
    print("Biryani")
elif Day == "Wednesday" or Day == "wednesday":
    print("Daal Chawal/Roti")
elif Day == "Thursday" or Day == "thursday":
    print("Aloo k parathae")
elif Day == "Friday" or Day == "friday":
    print("Nihari")
elif Day == "Saturday" or Day == "saturday":
    print("Aloo Matar Qeema")
elif Day == "Sunday" or Day == "sunday":
    print("Chicken Manchurian")
else:
    print ("Invalid Input")
```

What's the output with the following input?

Day = Friday	Day = funday	Day = monday

Program to check positive or negative number

1

```
# program to check positive or negative number

Number = float(input("Enter a number: "))
if (Number >= 0):
    if (Number == 0):
        print("The number is zero")
    else:
        print("Positive Number")
else:
    print ("Negative Number")
```

What's the output with the following input?

Number = 10000	Number = 0	Number = -90

Program to write numbers in a range with mentioned steps

1

```
start = int(input("Enter a number for start point: "))
stop = int(input("Enter a number for stop point: ")) + 1
step = int(input("Enter a number as a step in counting: "))

for i in range(start, stop, step):
    print (i)
```

What's the output with the following input?

Start = 0 Stop = 100 Step = 10	Start = 100 Stop = 0 Step = -10	Start = -100 Stop = -200 Step = -10

Program to write tables

1

```
# program to write tables

Number = int(input("Enter a number: "))

for i in range(11):
    print (Number, ' X ', i , ' = ', Number*i )
```

What's the output with the following input?

Number = 5	Number = 12	Number = 73

Exercise

2 x 10

1. Write a program to find the volume of cylinder (the cylinder's volume is $\pi r^2 h$)
2. Write a program that separately prints even numbers and odd numbers from a range specified by the user
3. Write a program to calculate sales tax on an item sold. Consider the following information:
Price of one packet of jellies = 10 rs
Price of one packet of juice = 50 rs
Price of one bag of chips = 40 rs
Price of one cup of ice-cream = 60 rs
Price of one bar of chocolate = 35 rs

Sales Tax = 5% of the item price

4. Write a program to calculate grade of each student. Consider the following information;
 - a. Grade A for students who obtain marks greater than or equal to 85%
 - b. Grade B when marks are greater than or equal to 75%
 - c. Grade C when marks are greater than or equal to 65%
 - d. Grade D when marks are greater than or equal to 55%
 - e. Grade E when marks are greater than or equal to 45%
 - f. Grade F when marks are below than 45%
 - g. Invalid marks if marks are greater than 100 or less than 0
 - h. Also adjust A+, B+, C+, D+, E+ grades

5. Serendipity Booksellers has a book club that awards points to its customers based on the number of books purchased each month.

The points are awarded as follows:

- If a customer purchases 0 books, he or she earns 0 points.
- If a customer purchases 1 book, he or she earns 5 points.
- If a customer purchases 2 books, he or she earns 15 points.
- If a customer purchases 3 books, he or she earns 30 points.
- If a customer purchases 4 or more books, he or she earns 60 points.

Write a program that asks the user to enter the number of books that he or she has purchased this month and then displays the number of points awarded.

6. Write a Menu driven program that outputs the following information about the shape:
 - a. For a rectangle, it outputs the area and perimeter
 - b. For a circle, it outputs the area and circumference
 - c. For a cylinder, it outputs the volume and surface area
7. Write a program to add numbers as much as the user wants. Hint: use while loop
8. This program displays the numbers 1 through 10 and their squares. Hint use for loop
9. Write a C++ Program that read an alphabet (e.g. a,b,c,d,...z) and display whether the input alphabet is a vowel (i.e. a, e, i, o, u) or consonant.
10. Consider the following program and output

```

start = int(input("Enter a number for start point: "))
stop = int(input("Enter a number for stop point: "))
step = int(input("Enter a number as a step in counting: "))
# outer loop to handle number of rows
# 10 in this case
for i in range(start, stop, step):

    # inner loop to handle number of columns
    # values changing acc. to outer loop
    for j in range(0, i):
        # printing stars
        print("* ", end="")

    # ending line after each row
    print("\n")

```

Enter a number for start point: 0
 Enter a number for stop point: 10
 Enter a number as a step in counting: 1

```

*
* *
* * *
* * * *
* * * * *
* * * * * *
* * * * * * *
* * * * * * * *
* * * * * * * * *

```

What should be the values of start, stop and step variables to draw the following pattern?

```

* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
*

```